Lovely Tech POS System

Final Requirements & Workflow Baseline — v2.0

FINALIZED IMPLEMENTATION BASELINE

Consolidated developer hand-over specification. Locked business rules are stated as implementation requirements; historical source material was reviewed for terminology and repair-form fields.

01 Project Overview

• System name: Lovely Tech POS System.

• Business scope: mobile phone sales, repairs, parts, accessories; laptops; computer repair, parts and accessories.

• Offline-first, branch-aware architecture; Internet is not required for core LAN operation.

• Initial roles: Admin/Owner and Cashier/Staff. Future Manager/custom roles are supported by the permission architecture.

• UI theme: pure red + black, glass-morphism/blur, shadows, optional lightweight animation, separate tabs, suitable for older laptops.

• Core rule: each transaction is entered once and its stock, FIFO, payment, warranty and audit effects commit atomically.

02 Roles & Security

• Cashier: sales, own completed-sales/reprint access, basic item entry/edit, sale-time price, permitted discounts, repair create/update/cancel, basic warranty claim, purchase entry without cost, stock/report view, day start/end/shift end, expense/other-income entry pending approval.

• Cashier never sees buy cost, supplier, FIFO unit cost, profit, internal pricing rationale or detailed backup/restore information.

• Admin has full control of costs, suppliers, stock adjustments, returns/exchanges, warranty decisions, approvals, users, permissions, reports, audit, backup/restore and settings.

• Permission checks are enforced in both UI and server/database layer.

• Admin auto-lock default 5 minutes; Cashier 1 hour. Repeated wrong passwords cause temporary lockout.

• Cashier password reset requires Admin verification and forced new password. Admin recovery uses one-time offline Recovery Code; recovery is audited.

• Critical Admin actions require re-authentication where configured: restore, critical void/correction, permission changes, stock adjustment, damage/loss approval, purchase-cost correction and Day End reopen.

• Audit log is immutable to normal users and Admin-only.

03 UI / Navigation

• Separate tabs/screens: Dashboard, Sales/POS, Repair, Warranty, Stock, Purchases, Expenses, Cash/Day Start-End, Reports and Admin.

• Color meaning: green success, yellow warning, red danger/error, blue/neutral information/history. Never rely on color alone.

• Search by item code, description, brand, model, barcode, category and IMEI/serial where applicable. In-stock first; out-of-stock below; negative stock visibly warned.

• Double-click/double-save protection and processing state are mandatory.

04 Item Master

• Sequential Item Code such as LVT0001; deleted codes are never reused. Imported existing codes must be preserved.

• Required basics: Item Code, Description, Category 1. Category 2 and Item Types are dynamic/admin-configurable.

• Multiple barcodes are supported. Manufacturer/custom barcodes map to the item.

• Master sale price is optional. Cashier/Admin may enter custom sale price at sale time; historical sale lines store actual price used.

• Phones/serialized items require unique IMEI/serial identity; duplicate identity sale is blocked.

• Repair parts can be compatible across models. Displays support quality/variant such as Original, High Quality, Phone Removed and Copy; displays use batch/quantity, not serial tracking.

• Default warranty is copied into the sale-time warranty record; later master changes never rewrite historical warranty.

05 Inventory & FIFO

• Stock is movement-based: purchase, sale, return, repair part usage/return, damage/loss, internal use and physical adjustment create traceable movements.

• Purchase lots are separate by date and cost. Exact FIFO allocation determines item cost and sales/repair profit.

• Negative stock is allowed with warning; sales are not blocked solely by insufficient stock.

• Later purchase retains full purchased lot quantity/cost. Earlier negative consumption is reconciled against later real purchase cost by deterministic FIFO reconciliation.

• Physical stock corrections are Admin-only stock movements with mandatory reason; positive corrections use the latest appropriate purchase-lot cost.

• Purchase cost, quantity or date corrections trigger affected FIFO/profit recalculation and audit.

• Returned stock preserves original cost lineage sufficiently for future FIFO.

06 Sales

• Flow: New Sale → search/scan → cart → quantity → price → discount → total → payment → document type → complete.

• Invoice numbers are sequential and never reused after void.

• Payment methods: Cash, Card, Online Transfer; split payment supported.

• Cashier can type sale price. Customer-facing documents show final selling price only; no buy cost/profit/internal rationale.

• Cashier discount is limited by user-specific configured limit; excess requires Admin approval.

• Customer details are optional. Credit sale is Admin-only and tracks total, paid and outstanding.

• Sale remains completed even if printer fails; reprint is a document event, not a new transaction.

07 Returns & Exchange

• Normal return window: 7 days; beyond configured Cashier authority is Admin-only. Partial returns supported.

• Refund is automatic; Admin may override with audit.

• Return charge is manually entered, has no fixed maximum, is separate Return Charge Income and is included in actual profit.

• Good return restocks; damaged return goes to Damage/Loss.

• Exchange links one Return and one completely separate New Sale. Customer first receives the old-item refund, then pays the full new-sale amount.

• Example: old 1000, charge 200 → refund 800; new item 1500 → customer pays 1500; net drawer +700; return-charge income 200.

• Cashier may mark exchange damage; Admin finalizes. If resellable, Admin reverses damage through audited restock.

• Important records are never hard-deleted.

08 Purchases

• Cashier purchase entry has no cost field. Basic invoice/quantities can be saved and stock increases immediately.

• Admin has private Purchase Cost Pending. Admin can edit lines, quantities, date/reference and item-by-item cost.

• Purchase payment method is not asked or stored. Supplier is optional/Admin-side.

• Completed costs become immutable purchase lots; corrections use controlled correction plus audit.

09 Repairs

• Repair IDs: REP-000001 style. Statuses: RECEIVED, CHECKING, WAITING FOR CUSTOMER, WAITING FOR PART, REPAIRING, READY, DELIVERED, RETURNED TO CUSTOMER, CANCELLED.

• Repair form supports the uploaded shop PDF fields: Device Brand, Given Accessories, Special Note, Model, Cust. Name, Contact No., Date In, Job No., Hand Over Date, Description/Amount, Repair Invoice, Advance Payment, Total, Reported Fault, Estimated Amount, Warranty Period and Authorized By.

• Estimate can differ from final amount. Labour is separate and labour-only repair is supported.

• Parts deduct FIFO stock. Removing good/resalable part restores its original FIFO lot; removing damaged part does not restore stock and enters Damage Review Pending for Admin.

• Advance + final payment are supported with Cash/Card/Online and split allocations. Final may be partial; outstanding remains. Delivery with outstanding requires Admin approval.

• Repair cancellation returns unused/good parts and sends damaged parts to Damage/Loss. Historical payments require explicit reversal/refund, never deletion.

• Repair warranty is separate from product warranty.

10 Warranty

• Product warranty starts on sale date; exact expiry/status stored. Sale-time override allowed.

• Warranty types: NONE, calendar DAYS, calendar MONTHS, CHECKING ONLY and controlled CUSTOM rules.

• Warranty claim is separate from repair. Cashier creates claim; Admin decides Supplier Warranty Claim or Damage/Loss.

• Accepted supplier replacement enters stock as WARRANTY_REPLACEMENT, not a purchase. Original claim/cost history remains linked.

• Supplier rejection: shop-held item → Damage/Loss; customer-kept item → no stock/profit write-off.

• Manual/handwritten invoice is captured inside normal sale by selecting Manual and recording the handwritten reference. Separate Manual Warranty Registration is exception-only for legacy/external/late cases.

• Lost bill recovery searches invoice/manual reference/IMEI/serial/customer/item/date. Reprint or duplicate never creates a second sale/warranty.

11 Cash / Day End / Shift End

• Day Start counts 5000, 2000, 1000, 500, 100, 50, 20 denominations. Opening correction Admin-only.

• Cash Drawer = opening cash + drawer-relevant cash inflows – refunds/outflows + Day-End Reload Total.

• Card/Online do not enter physical drawer.

• Cash In Hands = all approved shop inflows – all shop outflows, regardless of payment method. Purchases/expenses affect Cash In Hands but, under the locked rule, do not automatically reduce Day-End drawer.

• Other Cash In is an optional separate cash transaction, immediately updating Current Cash Balance and not automatically linking to main sales/stock/FIFO/main profit.

• Reload is entered once at Day End by Dialog/Airtel/Mobitel/Hutch. Cash-only; no live Current Cash Balance update; included in Day-End expected; brand-wise reporting and optional temporary estimated profit, not main profit.

• Day End shows expected, actual denomination total, difference, mandatory reason and optional note. Admin may reopen.

• Shift End is distinct from Day End. After Cashier Shift End, Admin has about one hour to choose Continue Today's Business; otherwise the Business Day auto-closes. Admin can later reopen by permission.

• Missed/backdated sale preserves original transaction date plus entry timestamp/user in audit; reports recalculate.

12 Expenses / Other Income

• Expense categories: Electricity, Rent, Internet, Transport, Repair Tools, Packing, Tea/Food, Other, Salaries; Admin can add categories.

• Cashier expense/other-income entries are Pending Approval until Admin approval. Pending items are excluded from final Profit and Cash In Hands.

• Expenses affect Cash In Hands but do not automatically alter Day-End drawer under the locked rule.

• Admin corrections preserve history.

13 Customers / Credit / Settlements

• Customer details are optional; history covers sales, repairs, warranty claims and bills.

• Credit sale is Admin-only. Revenue/profit may exist before cash collection.

• Later credit settlement is a payment-history event, not new revenue/profit. Store date/time, amount, method, previous outstanding and remaining balance.

• Repair balance settlement follows the same rule. Optional proof/receipt can be generated and reprinted when requested.

14 Reports / Targets

• Admin reports: Daily, Monthly, Yearly, Custom Date Range; PDF/Excel export.

• Separate Sales, Profit, Stock, Purchase, Repair, Warranty, Cash Drawer and Cash In Hands reports.

• Sales totals show Cash, Card and Online separately plus combined total.

• Monthly reports include previous-month comparison charts.

• Targets: Sales Amount, Item Sales Profit, Repair Income, Repair Profit, Bills Count, Items Sold Count. Cashier must not see numeric profit target.

• Progress may exceed 100%; visual bar stops at 100% while percentage continues. History is saved.

15 Backup / Restore

• Google Drive is backup/recovery, not live database sync.

• Backup stages: create → encrypt → upload → verify → verified. Failed verification is not a restore point.

• Cashier Shift End and Day End require backup. If network fails, Retry; if still unavailable, use a separate configured local/USB/external/network target, verify it, then allow close. Later queued backup syncs to Google Drive.

• Cashier sees Backup Status + Last Backup Time only. Admin sees verification and restore points.

• Restore is Admin-only: safety backup → explicit confirmation → restore verified point → integrity checks → audit.

• Interrupted backup keeps previous verified point valid. Interrupted restore is never reported successful. Split-brain databases and blind merging are prohibited.

16 Audit / Financial Rules

• Audit sales, edits, claims, discount limits, purchase corrections, stock adjustments, voids, returns/exchanges, day end, approvals, permissions, settlements, payment corrections, supplier decisions, damage/restock reversals, backup/restore, manual bills/reprints and security events.

• Corrections store old value → new value, user, time, reason and reference transaction.

• Profit = Sales Profit + Repair Profit + Other Income + Return Charge Income − Expenses − Damage/Loss and other applicable losses.

• Sales Profit uses actual sale revenue minus exact FIFO cost. Payment method does not change profit.

• Cash In Hands is separate from Profit and Cash Drawer. Credit/repair settlements are cash movements only. Pending approvals are excluded from final financials.

17 Failure / Validation Rules

• Sale, Return, Exchange, Repair, Purchase and Payment are atomic: all effects commit or all roll back.

• Unsaved drafts cannot change stock/payment/FIFO/profit.

• Idempotency/unique transaction IDs protect against retry after crash or double-click.

• Power failure during commit leaves either a complete transaction or none.

• Concurrent edits use version/stale-record protection; Admin priority applies where specified.

• Duplicate IMEI/serial is blocked. Printer failure never rolls back completed sale.

• Hard deletion of important financial, stock or warranty records is prohibited.